

MTL 704 - 18th April

Rosen Projection Gradient method

min $f(x)$

ST $x \in S = \left\{ x \in \mathbb{R}^n : \begin{array}{l} A_1 x \leq b_1 \\ A_2 x = b_2 \end{array} \right\}$

$$A_1 \rightarrow m_1 \times n$$

$$A_2 \rightarrow m_2 \times n$$

$$b_1 \rightarrow m_1 \times 1$$

$$b_2 \rightarrow m_2 \times 1$$

S is non empty and compact
↳ this guarantees existence of solution.

initial pt $x_0 \in S$

To find such an x_0 , we can solve a LP

$$\min \quad 0^T x$$

$$\text{ST} \quad A_1 x \leq b_1$$

$$A_2 x = b_2$$

let x_k be the current iterate in S

$$I(x_k) = \{\text{active indices @ } x_k\}$$

let $|I(x_k)| = q$ then $m_1 \leq q \leq m_1 + m_2$

$$A_q x_k = b_q$$

$$A_q = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}_{q \times n}$$

$$b_q = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Tangent plane @ x_k created by active indices

$$M_k = \left\{ x \in \mathbb{R}^n \mid A_q x = 0 \right\} \quad \leftarrow \begin{matrix} \text{subspace} \\ \text{of } \mathbb{R}^n \end{matrix}$$

project $-\nabla f(x_k)$ on M_k
take $d_k = P_k(-\nabla f(x_k))$

$$A_q : q \times n$$

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^q, \quad T(x) = A_q x$$

$$M_k = \text{Null space of } A_q \\ \equiv \text{Null space of } T$$

$$\text{let } N_k = \text{range space of } A_q \\ = \{ A_q^T y : y \in \mathbb{R}^q \}$$

$$\text{rank}(A_q) = \dim(N_k)$$

By rank-nullity theorem

$$\mathbb{R}^n = M \oplus N$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 range space null space
 direct sum

$$-\nabla f(x_k) \in \mathbb{R}^n$$

$$\therefore -\nabla f(x_k) = d_k + A_q^t y_k$$

$$\text{for some } y_k \in \mathbb{R}^q$$

d_k and y_k are unique

$$A_q d_k = 0$$

$$A_q^t y_k = -(\nabla f(x_k) + d_k)$$

$$\Rightarrow \underbrace{A_q A_q^t}_{\substack{q \times q \\ \text{matrix}}} y_k = -A_q (\nabla f(x_k) + d_k)$$

$$= -A_q \nabla f(x_k)$$

full rank $\Rightarrow$ invertible

$$y_k = \underbrace{-(A_q A_q^t)^{-1} A_q}_{n \times 1} \nabla f(x_k)$$

$$y_k \in \mathbb{R}^q$$

$$\Rightarrow -\nabla f(x_k) = d_k + A_q^t (-(A_q A_q^t)^{-1} A_q \nabla f(x_k))$$

$$\Rightarrow d_k = -\nabla f(x_k) + A_q^t ((A_q A_q^t)^{-1} A_q \nabla f(x_k))$$

$$\Rightarrow d_k = \underbrace{\left[I - A_q^t (A_q A_q^t)^{-1} A_q \right]}_{n \times n} \underbrace{(-\nabla f(x_k))}_{n \times 1}$$

$$\Rightarrow d_k = \underbrace{P_k}_{n \times 1} \cdot (-\nabla f(x_k))$$

projection matrix

$\rightarrow P_k^t = P_k$
 $\rightarrow P_k$ should be symmetric

~~$d_k = 0$~~

$$\rightarrow P_k^2 = P_k$$

$$\begin{aligned} P_k^t &= I - (A_q^t (A_q A_q^t)^{-1} A_q)^t \\ &= I - A_q^t \left((A_q A_q^t)^{-1} \right)^t A_q = P_k \end{aligned}$$

$$\left((A_q A_q^t)^t \right)^{-1} = (A_q A_q^t)^{-1}$$

$\therefore P_K$ is symmetric

$$\begin{aligned} P_K^2 &= \cancel{I \otimes A} \\ &= (I - B)(I - B) \\ &= I - 2B + B^2 \end{aligned}$$

so we have to show

$$-2B + B^2 = -B$$

$$\text{or } B^2 = B$$

$$\begin{aligned} B &= A_q^t (A_q A_q^t)^{-1} A_q \rightarrow I \\ B^2 &= A_q^t (A_q A_q^t)^{-1} A_q \times A_q^t (A_q A_q^t)^{-1} A_q \end{aligned}$$

$$= A_q^t (A_q A_q^t)^{-1} A_q = B$$

$$\therefore \boxed{P_K^2 = P_K}$$

Now choose α_k st

$$x_k + \alpha_k d_k \in S$$

$$\min_{\alpha_1 \leq \alpha \leq \alpha_2} \phi(\alpha) = f(x_k + \alpha d_k)$$

$$A_1 (x_k + \alpha d_k) \leq b_1$$

$$A_2 (x_k + \alpha d_k) = b_2$$

$$x_k \in S$$

$$\Rightarrow A_1 x_k \leq b_1$$

Part of A_1 (if any)
which is in $I(x_k)$,
we have

$$A_1^1 d_k = 0$$

$\therefore$ Any α will work

$$\text{only for } A_1^2 x_k < b_1^2$$

$\hookrightarrow$ indices which are inactive.

$$A_1^2 x_k + \alpha A_1^2 d_k \leq b_1$$

$$\Rightarrow \alpha A_1^2 d_k \leq \underbrace{b_1 - A_1^2 x_k}_{> 0}$$

$(m_1 + m_2 - q)$ $\hookrightarrow$ vector $\hookrightarrow$ we or -ve components (no zero)
inequalities

satisfied $\forall \alpha$

because $x_k \in S$

$$A_2 x_k = b_2$$

and A_2 was
part of Active
indices

$$\therefore A_2 d_k = 0$$

we have to find
 α restriction

$$\alpha \leq \left(\frac{b_1^2 - A_1^2 x_k}{A_1^2 d_k} \right) \text{ its component of } A_1^2 d_k > 0$$

this gives α_2

sym: for -ve component
it gives α_1

$$x_{k+1} = x_k + \alpha_k d_k$$

Stopping $\Rightarrow$ when KKT condⁿ satisfied.
because convex problem.

Suppose we compute projection:

$$P_k(\nabla f(x_k)) = 0 = d_k$$

$$\mathbb{R}^n = M \oplus N$$

$\uparrow$ $\uparrow$
 $-\nabla f(x_k)$ d_k

$$-\nabla f(x_k) = d_k + A_q^t y_k^*$$

$$\Rightarrow \nabla f(x_k) + A_q^t y_k = 0$$

$$\hookrightarrow \nabla L = 0$$

Active index

$$(P) \quad \min f(x)$$

$$s.t. \quad A_1 x \leq b_1 \quad \rightarrow \lambda$$

$$A_2 x = b_2 \quad \rightarrow \mu$$

$$L = f(x) + \lambda(A_1 x - b_1) + \mu(A_2 x - b_2)$$

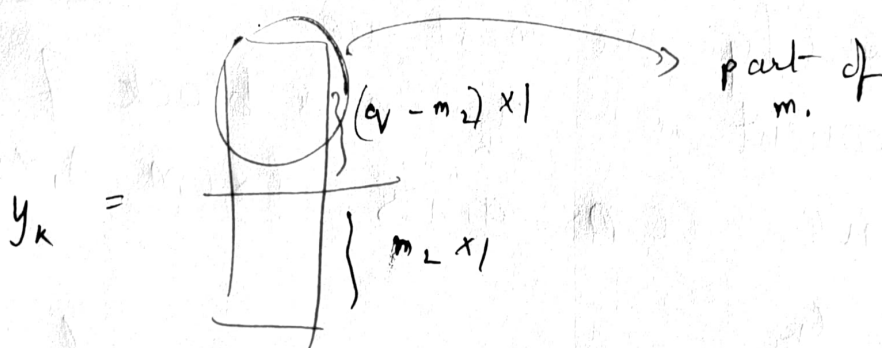
$$\lambda(A_1 x - b_1) = 0$$

$$\lambda \geq 0$$

μ : unrestricted

$$\mu(A_2 x - b_2) = 0$$

$$\nabla L = \nabla f(x) + A_1^T \lambda + A_2^T \mu = 0$$



If $d_k = 0$ and component of y_k corresponding to active indexes from $A_1 x \leq b_1$ @ x_k are non-ve then x_k is KKT pt of (P) hence stop.

If $d_k = 0$ but there is atleast 1 component of y_k corresponding to active constraints $A_i x \leq b_i$

(a) x_k is strictly -ve

then we do not get KKT pt.

~~But in CP~~

$$x_{k+1} = x_k$$

We identify the most -ve component of y_k in a block of active constraints from $A_i x \leq b_i$

(a) x_k

Suppose this is j th component.

Delete the j th row in A_k to get a new matrix $A_{\hat{k}}$, $\hat{k} < n$.

Compute new p_k with $A_{\hat{k}}$

If $d_k = 0$ and we do not have
KKT pt $\mathcal{J}_f(p)$ and we have
only one row in A_q which also
ask for deletion

So computing P_k is impossible

so now take $d_k = -\nabla f(x_k)$

(there is no A_q)
 $\therefore P_k = \underline{I}$